THE HIGHWAY EXAMINATIONS BOARD

PRIMARY SEVEN PRE-MOCK SET 1 MARKING GUIDE

2025

INTEGRATED SCIENCE

Time Allowed: 2 hours 15 minutes

Index No.	**Random No.**						**Personal No.**		

Candidate's Name: .....

Candidate's Signature: .....

Read the following instructions carefully:

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two sections **A** and **B**. Section **A** has **40 questions** and **Section B** has **15 questions**. This paper has **15 pages** printed altogether.
3. Answer **all** questions. All the working for both sections **A** and **B** must be shown in the spaces provided.
4. **All** answer **must** be written using a **blue** or **black** ball point pen or ink. Any work written in pencil will **NOT** be marked.
5. Unnecessary **changes** in your work and handwriting that cannot be easily ready may lead to **loss of marks**.
6. Do not fill anything in the box indicated: **"FOR EXAMINERS' USE ONLY"** and the boxes inside the question paper.

FOR EXAMINERS' USE ONLY		
QN. NO.	**MARKS**	**EXR'S NO**
1 - 10		
11 - 20		
21 - 30		
31 – 40		
41 – 43		
44 – 46		
47 – 49		
50 – 52		
53 – 55		
TOTAL		

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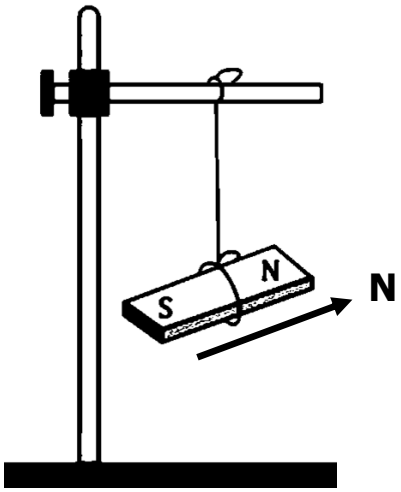
SECTION A: 40 MARKS

Questions **1** to **40** carry **one mark** each

1. Name the type of latrine that separates faeces from urine.
   - ✓ **Ecosan latrine**
2. Identify the source of kinetic energy used in the production of electricity at a geothermal power station.
   - ✓ **Hot springs**
3. Give the meaning of the term soil erosion.
   - ✓ **Is the removal of topsoil.**

     Reject:
     - ❖ **is the removal of top soil by its agents.**
     - ❖ **is the gradual removal of top soil.**
4. Identify the role of the liver as an excretory organ.
   - ✓ **It removes bile pigments from the body.**

The diagram below shows a property of magnets. Use it to answer questions **5** and **6**.



5. Name the property of magnets shown above.
   - ✓ **A freely suspended bar magnet rests in the north – south direction.**
6. Give **one** way in which the property of magnets above is useful to pilots.
   - ✓ **Helps pilots to find direction of places using magnetic compass.**
   - ✓ **Helps to show direction of flight.**

7. Apart from adolescent girls, give **one** other group of people who are eligible to receive Tetanus Toxoid (T.T) vaccine during immunization.
   - ✓ **Pregnant mothers**
8. Mention any **one** human activity that can lead to soil exhaustion.
   - ✓ **Monocropping.**
   - ✓ **Overuse of artificial fertilisers.**
   - ✓ **Dumping harmful materials on soil.**
9. Name the germ that cause bilharzia.
   - ✓ **Worms**

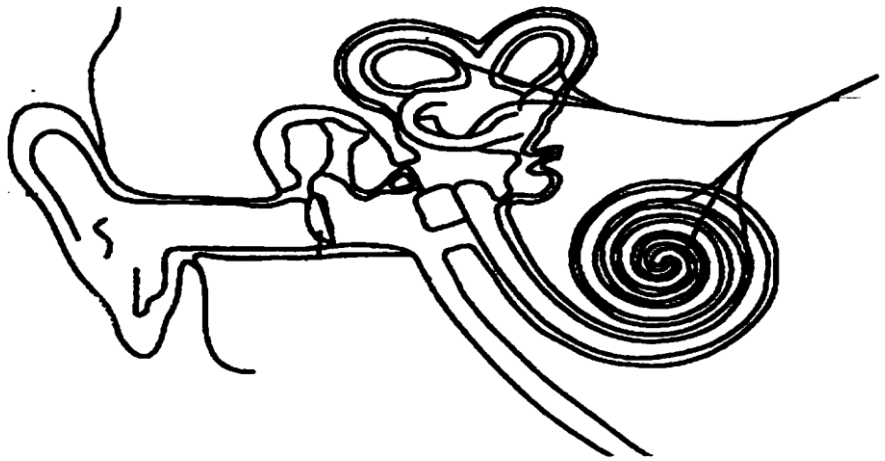
     Reject:
   - ❖ **Bilharziasis // Blood flukes**
10. Identify the group of living things to which a butterfly belongs.
    - ✓ **Animals.**

      Reject:
    - ❖ **Invertebrates //Arthropods**
    - ❖ **Insects**
11. Give any one exotic breed of rabbits kept in your community.
    - ✓ **Dutch rabbits**
    - ✓ **Angora rabbits**
    - ✓ **Rex rabbits**
    - ✓ **Californian rabbits**
    - ✓ **Chinchilla rabbits**
    - ✓ **New Zealand white**
    - ✓ **Lop rabbits**

    Mnemonic: **DR. CLANC**
12. Identify any **one** example of non – living resources.
    - ✓ **Soil**
    - ✓ **Water**
    - ✓ **Minerals**
    - ✓ **Air**.
    - ✓ **Fossil fuels**
13. Mention any **one** example of common accidents on roads.
    - ✓ **Drowning**
    - ✓ **Near drowning**
    - ✓ **Road traffic accidents**

14. Give any **one** way in which fungi are harmful to people in the environment.
    - ✓ **Moulds make food to go bad.**
    - ✓ **Toadstools are poisonous when eaten.**
    - ✓ **Fungi cause food poisoning.**
    - ✓ **Pathogenic fungi cause diseases.**
15. State any **one** way in which friction can be reduced in machines.
    - ✓ **By using rollers.**
    - ✓ **By using ball bearings.**
    - ✓ **By lubricating moving parts.**
    - ✓ **By streamlining moving bodies.**

    The diagram below shows a human organ. Use it to answer questions **16** and **17.**



16. Name the organ shown above.
    - ✓ **Ear / Human ear**
17. Identify the part of an insect which performs the same function as the above human organ.
    - ✓ **Feelers / feeler**
    - ✓ **Antannae**

18. State any **one** condition that leads to swarming of bees.
    - ✓ **Over crowdedness of bees in the hive.**
    - ✓ **Development of the new queen bee.**
    - ✓ **Lack of flowers.**
    - ✓ **Lack of water.**
    - ✓ **Direct sunshine into the hive.**
    - ✓ **Too much noise around the hive.**
    - ✓ **Bad smell around the hive.**
    - ✓ **Leakage in the hive.**
    - ✓ **Attack by pests.**
19. Identify any **one** example of plant habitats.
    - ✓ **Garden**
    - ✓ **Wetlands**
    - ✓ **Rock places**
    - ✓ **Water**
    - ✓ **Dry areas**
20. Mention any **one** type of chicken.
    - ✓ **Layers**
    - ✓ **Dual purpose**
    - ✓ **Broilers**
21. State the danger of teenage pregnancy to girls.
    - ✓ **Leads to parental reject.**
    - ✓ **Leads to forced marriage.**
    - ✓ **Leads to school dropouts.**
    - ✓ **Leads to death due to attempted abortion.**
    - ✓ **Leads to difficulty in giving birth / Birth complication.**
22. Give any one reason why sheep farmers carryout docking.
    - ✓ **To prevent flystrikes.**
    - ✓ **To keep the sheep clean.**

      Reject**:**
    - ❖ **To ease mating**
23. State the cause of marasmus in children.
    - ✓ **Lack of all food nutrients in the diet**
    - ✓ **Lack of all food values in the diet.**

24. State how isolation of infected animals prevent viral diseases in cattle.
    - ✓ **It prevents close contact of infected animals with healthy ones.**
25. Give any **one** activity at home that requires the filtration method.
    - ✓ **When separating water from soil.**
    - ✓ **When separating juice from fruits.**
    - ✓ **When separating tea from tea leaves.**
    - ✓ **When preparing local salt.**
26. State any **one** way conducting health parades promote health at school.
    - ✓ **Encourages children to bathe.**
    - ✓ **Encourages children to wash clothes.**
    - ✓ **Encourages children to iron clothes.**
    - ✓ **Encourages children to wear clean clothes.**
27. Give any one characteristic of living things.
    - ✓ **They reproduce**
    - ✓ **They breathe**
    - ✓ **They eat / feed.**
    - ✓ **They grow.**

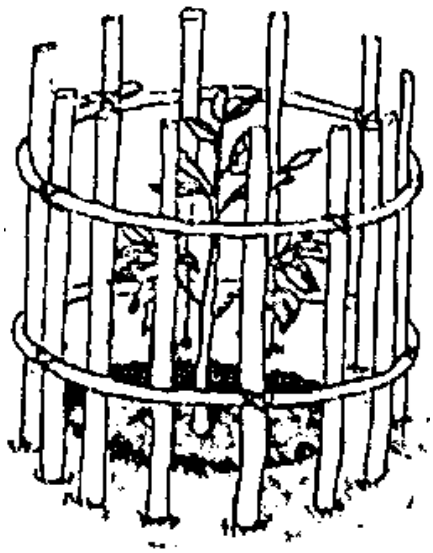
    Reject**:**
    - ❖ **They die**.
28. Identify one material used for keeping the body clean.
    - ✓ **Pumice stone**
    - ✓ **Soap**
    - ✓ **Razorblade**
    - ✓ **Bathing sponge**
    - ✓ **Comb**
    - ✓ **Handkerchief**
    - ✓ **Water**
    - ✓ **Nail clipper**
    - ✓ **Towel**
    - ✓ **Cotton swabs**
    - ✓ **Brush**

    Reject: **Ear buds**

29. Give any **one** advantage of breast feeding to the family.
    - ✓ **It is cheap in terms of money.**
    - ✓ **It requires less labour in feeding the baby.**
    - ✓ **It brings happiness to the family when the baby and mother are healthy.**
    - ✓ **Each baby in the family gets a healthier start of life.**
30. Mention any **one** way roughage help to prevent constipation.
    - ✓ **Roughage increase water absorption.**
    - ✓ **Roughage adds bulk in stool.**
    - ✓ **Roughage stimulates intestinal muscle contraction.**

      Reject:
    - ❖ **Roughage absorbs water from the stool.**
31. Give **one** way in which the use of a thermos flask is friendly to the environment.
    - ✓ **It reduces the rate at which the trees are cut for wood fuel to boil liquids.**
    - ✓ **It reduces pollution by reducing the use of disposed thermoplastics containers.**

    The diagram below shows a practice carried out in tree growing in agroforestry. Use it to answer questions **32** and **33**.



32. Identify the tree growing practice shown above.
    - ✓ **Tree fencing / fencing trees.**
33. State any one way in which the above tree growing practice is important in agroforestry.
    - ✓ **Protect trees from damage by animals and young children.**
    - ✓ **Protect trees from being destroyed by strong wind.**

34. State any **one** role played by the school in the control of antisocial behaviour in children.
    - ✓ **Providing equal treatment to all children.**
    - ✓ **Forming school clubs.**
    - ✓ **Engaging children in games and sports.**
    - ✓ **Designing school rules and regulations.**
    - ✓ **Setting penalties for wrong doers.**
    - ✓ **Organising guidance and counselling session.**
35. Identify any **one** way in which vaccines are administered into the body.
    - ✓ **Through injection.**
    - ✓ **Through drops in the mouth**
36. Apart from adventitious roots, give any one other type of roots.
    - ✓ **True roots**

      Reject:
      - ❖ **Tap roots/Prop roots/Aerial roots/Clasping roots /Buttress roots/Stilt roots/Storage roots/Fibrous roots**

      NB: We don't have primary and secondary roots as types of roots at primary level they are meant for higher level.
37. State **one** way in which sunlight is useful during photosynthesis.
    - ✓ **Provides energy needed by plants to make glucose / food.**

      Reject:
      - ❖ **Split water into hydrogen and oxygen.**
38. Give any **one** way the circulatory system benefits from the skeleton.
    - ✓ **Skeleton protects the heart.**
    - ✓ **Skeleton helps in the production of blood cells.**
39. Mention any **one** device used to store sound by electromagnetic method.
    - ✓ **Compact discs (CDs)**
    - ✓ **Hard discs**
40. State any one way of increasing the volume of blood in the human body.
    - ✓ **Through blood transfusion.**
    - ✓ **Taking iron supplement tablets.**
    - ✓ **Feeding on food rich in iron.**
    - ✓ **Drinking plenty of fruit juice and water regularly.**

SECTION B: 60 MARKS

Questions **41** to **55** carry four marks each.

41. (a) Give the meaning of the term balanced diet.
    - ✓ **Is a diet that contains all food values in their right amount.**

      Reject:
      - ❖ **A balanced diet is a meal which contains all food values in their right amount.**

    (b) Write down **two** reasons why it is important to feed on a balanced diet.
    - ✓ **To protect the body against malnutrition.**
    - ✓ **To promote proper body growth.**

    (c) State any **one** way in which feeding on vitamin **A** is important to adults.
    - ✓ **Helps in the proper functioning of the eye retina.**
    - ✓ **Prevents night blindness.**

42. (a) State any **two** ways in which germs can spread.
    - ✓ **Through drinking unboiled water.**
    - ✓ **Through body contact.**
    - ✓ **Through bites of dogs or insects.**
    - ✓ **Through inhaling contaminated air.**
    - ✓ **Through sharing skin cutting objects.**
    - ✓ **Through sharing clothes.**

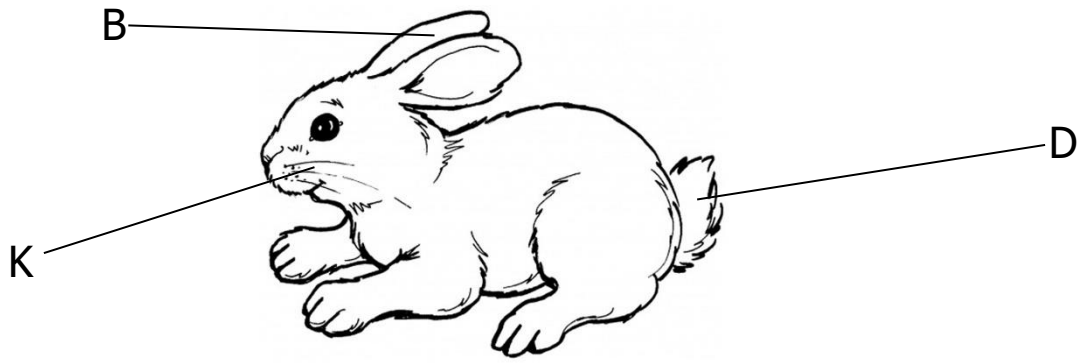
    (b) Mention any **two** ways in which the spread of germs can be controlled in an area.
    - ✓ **By covering leftover food properly.**
    - ✓ **By collecting and burning rubbish.**
    - ✓ **By washing hands with clean water and soap.**
    - ✓ **By drinking boiled water.**

43. (a) State any **two** qualities of a good stake when staking tomato plants.
    - ✓ **Should be strong.**
    - ✓ **Should not rot easily.**
    - ✓ **Should not be too close to the plant.**

(b) Identify any **two** reasons why farmers stake their tomato plants.

- ✓ **To ease weeding and harvesting.**
- ✓ **To enable plants get enough sunlight tomato food.**
- ✓ **To prevent ground pests from damaging fruits.**

44. The diagram below shows an animal commonly kept at home. Study and use it to answer questions that follow.



(a) Name the parts labelled;

(i) K - **Whiskers**

(ii) D - **Tail**

(b) Identify the part of the above animal which is handled when transporting it.

✓ **Part B** Reject**: Ear**

(c) State any **one** advantage of keeping the above animal over goats.

- ✓ **The above animal grow faster than goats.**
- ✓ **The above animal reproduce faster than goats.**
- ✓ **The above animal need less space than goats.**
- ✓ **The above animal is cheaper to buy than goats.**

45. The table below shows energy resources and their sources. Complete the table correctly.

Energy resources	**Sources**
(i) **wood fuel/ Bio fuel.** **Reject:** food and biogas	Plants
Uranium	(ii) **minerals**
(iii) **Animal energy** **Reject:** food and biogas	Animals
Wind energy	(iv) **Wind**

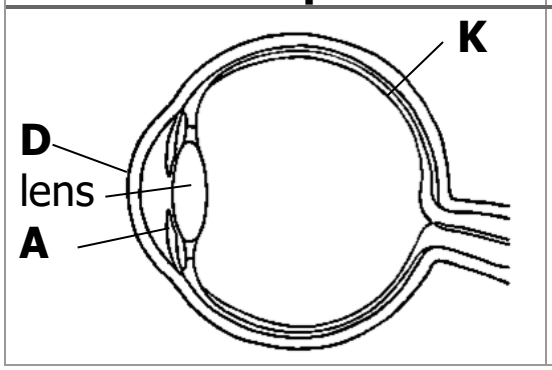
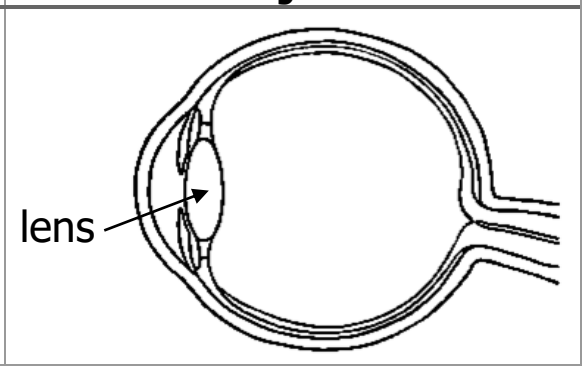
46. You are provided with the following materials.
Toothpaste, water, toothbrush. Describe the steps you would follow to brush the teeth.
    (i) Wet the toothbrush and apply small amount of toothpaste.
    (ii) Brush the teeth up and down.
    (iii) Brush the tongue to remove food remains and freshen the breath.
    (iv) Rinse the mouth and toothbrush with water.

47. (a) Apart from deep litter system, give any **two** other systems of keeping poultry.
    - ✓ **Free range system.**
    - ✓ **Battery system // cage system**
    - ✓ **Fold system.**

    Reject:
    - ❖ **Battery cage system.**

    (b) Explain **one** way in which perches control poultry vices in a deep litter system. (**2 marks**)
    - ✓ **Perches enable birds to do physical exercise hence preventing boredom.**

48. The diagram below is of a human eye. Study and use it to answer questions that follow.

Accept	Reject
	

(a) Name the parts marked;
D - **Pupil**
A - **Iris**

(b) State any **one** characteristic of the image formed at part **K**.
- ✓ **The images are real.**
- ✓ **The images are inverted.**
- ✓ **The images are diminished.**

(c) With the help of an arrow use letter **P** to show the lens.

49. (a) State any **one** way of caring for the following tuber crops;

(i) Cassava.

- ✓ **Regular weeding**
- ✓ **Pruning**
- ✓ **Spraying pesticides**
- ✓ **Earthing up**

(ii) Coco yams.

- ✓ **Weeding**
- ✓ **Earthing up**

(b) Mention any **two** ways of processing cassava.

- ✓ **By peeling**
- ✓ **Roasting**
- ✓ **Boiling**
- ✓ **Sun drying**
- ✓ **Deep frying**
- ✓ **Grating**
- ✓ **Slicing cassava**
- ✓ **Washing cassava**
- ✓ **Milling.**

50. (a) Identify any **two** properties of matter.

- ✓ **Matter has mass.**
- ✓ **Matter has volume.**
- ✓ **Matter occupies space.**

Reject:

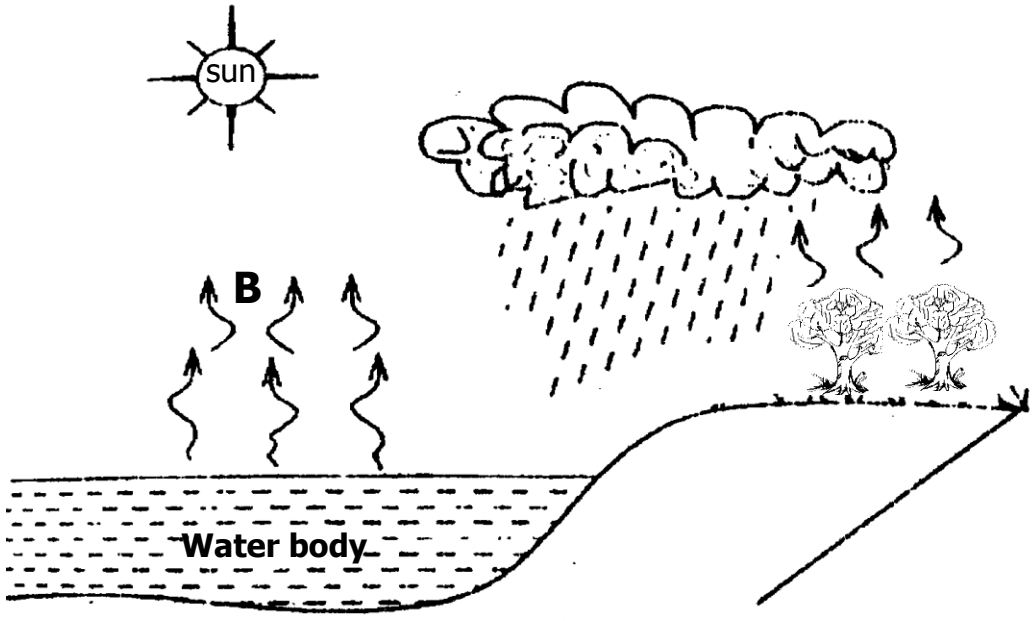
- ❖ **Matter has weight.**

(b) Name the state of matter in which heat travel fastest by;

(i) convection - **gases**

(ii) conduction - **solids**

51. The diagram below shows a water cycle. Study and use it to answer the questions that follow.



(a) Identify the process that is taking place at **B**.

✓ **Evaporatio**n.

(b) State any **one** danger of the following in the environment.

(i) Sun:

✓ **It increases temperature.**

✓ **It dries crops.**

(ii) Clouds:

✓ **They cause lightning which kills people.**

✓ **They cause heavy storms which spoil property and crops**

(c) Give **one** way in which plants are important in the above.

✓ **They add vapour to the atmosphere through transpiration.**

52. (a) Identify the nitrogenous wastes which is found in urine.

- ✓ **Urea**

Reject:

- ❖ **Uric acid**
- ❖ **Ammonia**

(b) State any **one** function of the kidney in the human body.

- ✓ **Produces urine / helps in urine production.**
- ✓ **Balances the amount of water needed in the body.**
- ✓ **Controls the level of salts in blood.**

Reject:

- ❖ **Filters blood**

(c) Give a reason why wetlands are referred to as the natural kidney of the earth.

- ✓ **Wetlands filter water before drained / poured into water lakes and rivers.**

(d) Mention any **one** way of keeping the excretory system healthy.

- ✓ **Feeding on a balanced diet.**
- ✓ **Doing regular physical body exercises.**
- ✓ **Going for regular medical check-ups.**

53. (a) Name the method used to find the volume of regular objects.

- ✓ **Displacement method.**

Reject:

- ❖ **Formula**
- ❖ **Volume = length X width X height**

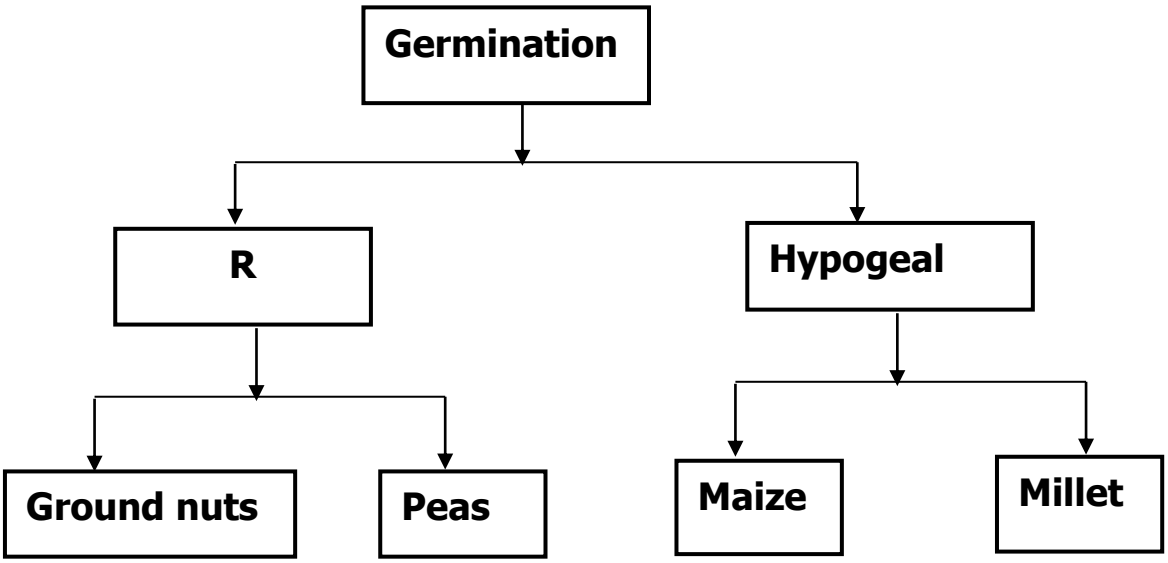
(b) A stone of density **8g/cc** and mass **160g** was lowered in a container full of water overflowed. Calculate the volume of water that over flowed.

$$\textbf{Volume} = \frac{\textbf{Mass}}{\textbf{Density}}$$

$$\textbf{Volume} = \frac{160g}{8g} = 80$$

Volume = 80cc

54. The table below shows different types of germination. Study it carefully and answer questions that follow.



(a) Name the type of germination marked **R**.

✓ **Epigeal germination.**

(b) Give a reason why groundnuts are grouped under the above type of germination.

✓ **The cotyledon comes above the ground.**

(c) Identify any **two** conditions needed for germination to take place.

✓ **Water**

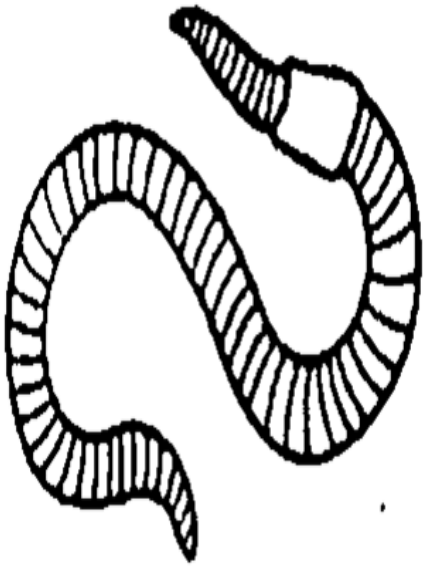
✓ **Oxygen**

✓ **Warmth**

Reject:

❖ **Moisture**

55. The table below shows different groups of worms. Use them to answer the questions that follow.

A	B
	

(a) Identify the group of worms marked **A**.

✓ **Segmented worms.**

(b) Give any **one** way of controlling worm **A** in humans.

✓ **Eating well cooked meat.**

✓ **Regular deworming.**

✓ **Eating proper roasted meat.**

✓ **Proper disposal of faeces.**

(c) State any **two** ways in which worm **B** is important in the environment.

✓ **Improves on soil fertility.**

✓ **Improves water drainage in the soil.**

✓ **Helps in the formation of humus.**

✓ **Aerates the soil by creating tunnels in the soil.**

Reject:

❖ **They die and form humus**